Efficiency Corrections Comparison

Current vs. Previous Results

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March 27, 2026

Methodology

Reconstruction Efficiency and Covariance

1. Mean Reconstruction Efficiency: Calculated as the arithmetic mean of event-by-event interpolated efficiencies for N events in a bin:

$$\bar{E}_{\text{reco}} = \frac{1}{N} \sum_{i=1}^{N} E_{\text{reco},i}$$

2. Correlated Error Propagation: Because efficiencies are interpolated from the same D_1 nodes, event uncertainties (σ_i) are correlated. We construct a covariance matrix:

$$Cov_{i,j} = \rho_{i,j} \sigma_i \sigma_j$$

- $\rho = 1$ if events share identical or adjacent D_1 interpolation nodes.
- $\rho = 0$ if events use entirely different nodes.

$$\sigma_{\bar{E}_{\text{reco}}} = \frac{1}{N} \sqrt{\sum^{N} \sum^{N} Cov_{i,j}}$$

Methodology

Final Efficiency Corrections

1. Hodoscope Mean Efficiency: Evaluated with purely independent, standard weighted uncertainty:

$$\sigma_{\bar{E}_{\text{hodo}}} = \frac{1}{N}\sqrt{\sum_{i=1}^{N} \sigma_{\text{hodo},i}^2}$$

2. Final Bin Efficiency: Computed as the product of the independent bin averages:

$$\bar{E}_{\text{final}} = \bar{E}_{\text{reco}} \times \bar{E}_{\text{hodo}}$$

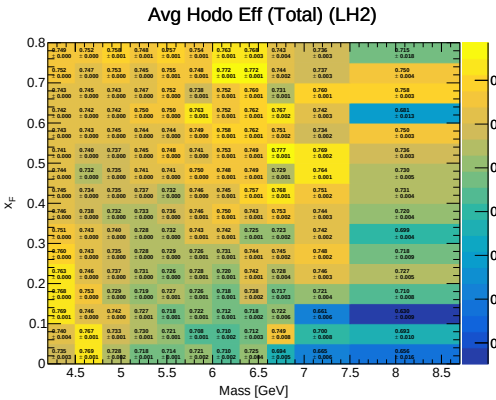
3. Final Propagated Uncertainty:

$$\sigma_{\bar{E}_{\text{final}}} = \sqrt{\left(\bar{E}_{\text{hodo}} \cdot \sigma_{\bar{E}_{\text{reco}}}\right)^2 + \left(\bar{E}_{\text{reco}} \cdot \sigma_{\bar{E}_{\text{hodo}}}\right)^2}$$

LH2 Efficiency Corrections

LH2 Target: Total Yield

Hodoscope Efficiency Corrections



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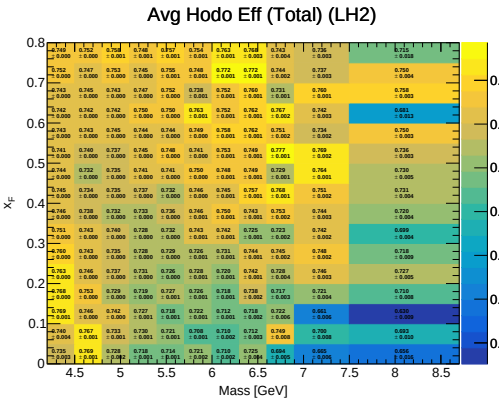
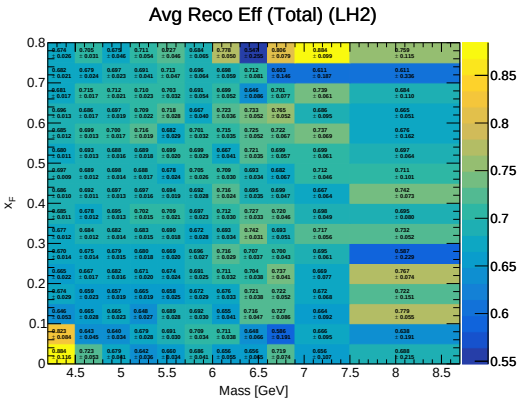


(b) Current

LH2 Target: Total Yield

Reconstruction Efficiency Corrections



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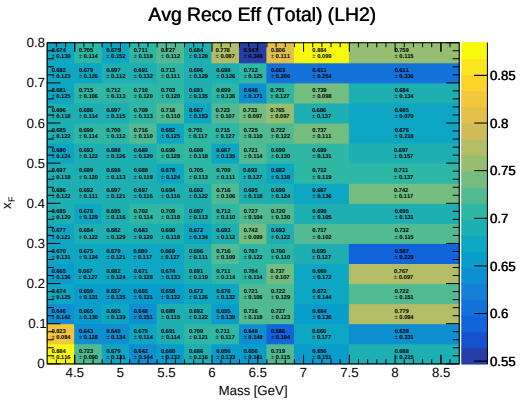
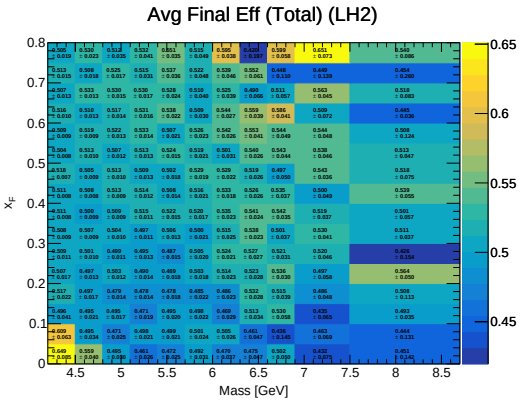


(b) Current

LH2 Target: Total Yield

Final Efficiency Corrections



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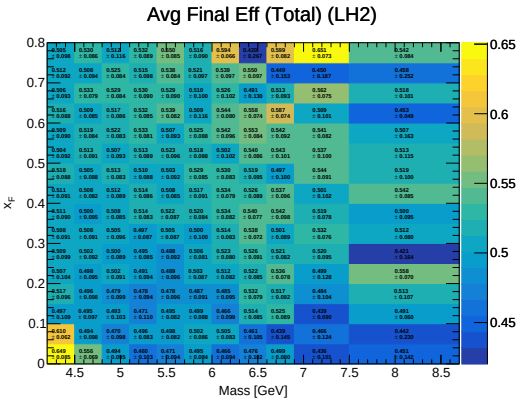

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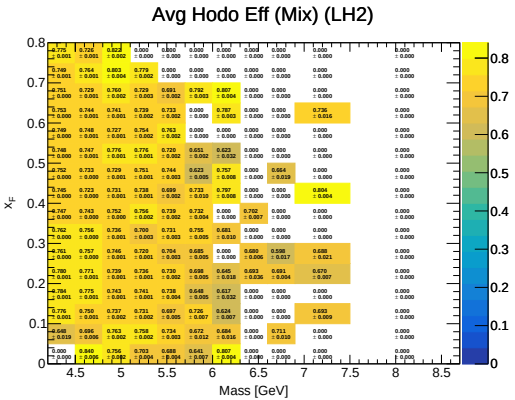
LH2 Target: Mix Yield

Hodoscope Efficiency Corrections



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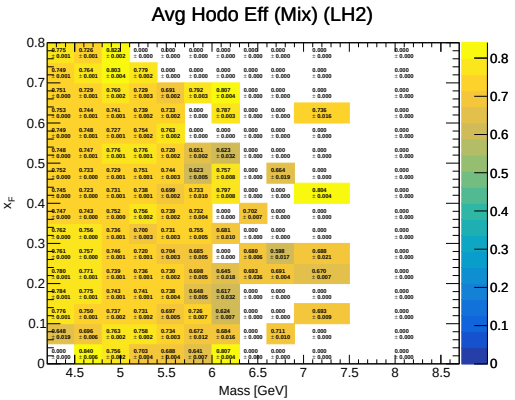

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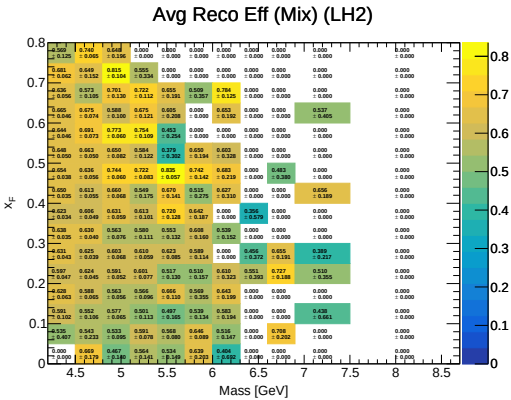
LH2 Target: Mix Yield

Reconstruction Efficiency Corrections



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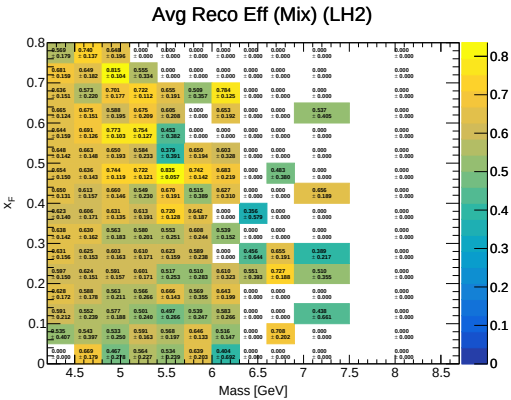

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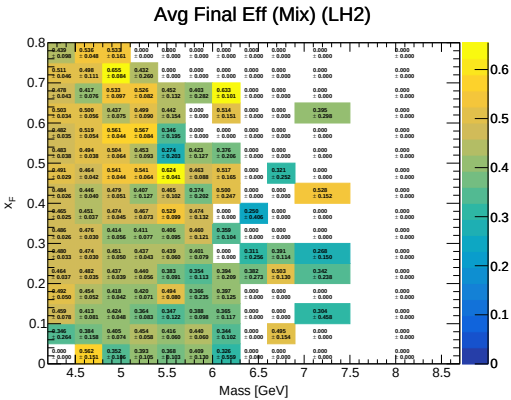
LH2 Target: Mix Yield

Final Efficiency Corrections



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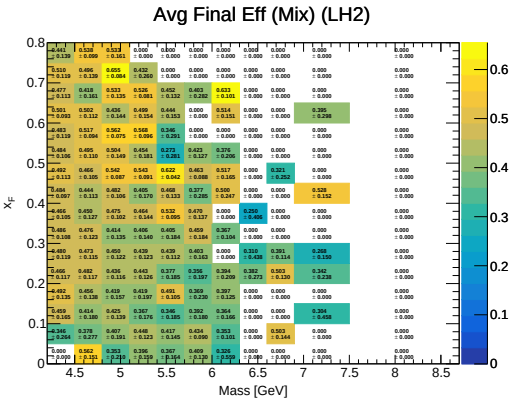
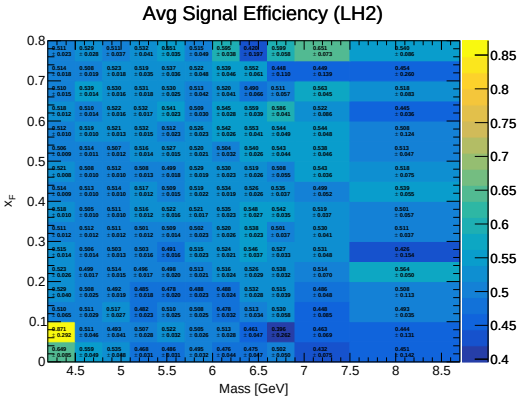


(b) Current

LH2 Target: Signal Efficiency

Final Corrections



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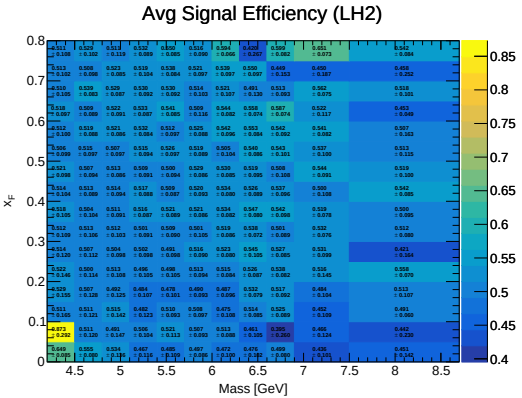

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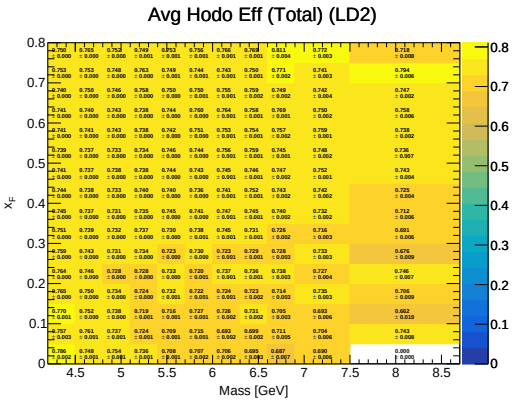
LD2 Efficiency Corrections

LD2 Target: Total Yield

Hodoscope Efficiency Corrections



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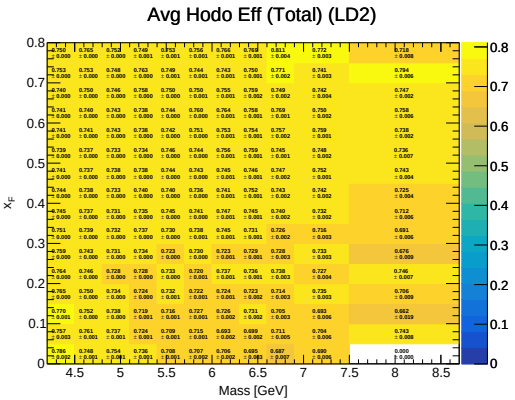

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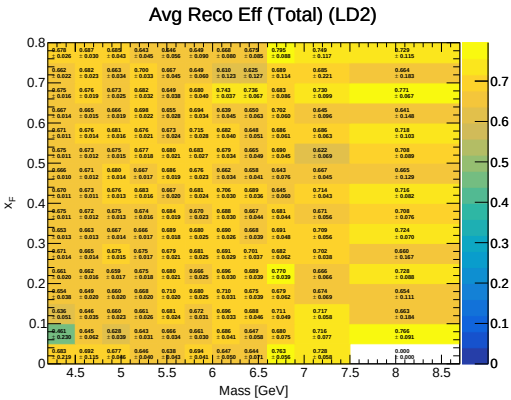
LD2 Target: Total Yield

Reconstruction Efficiency Corrections



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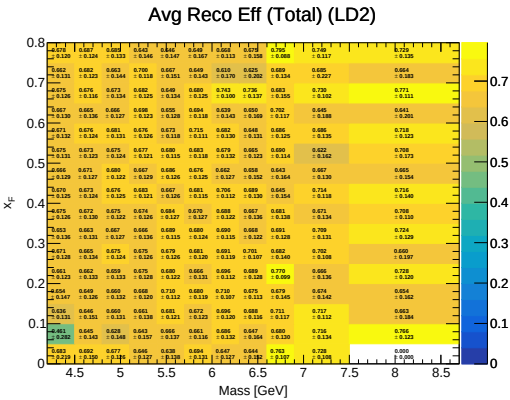

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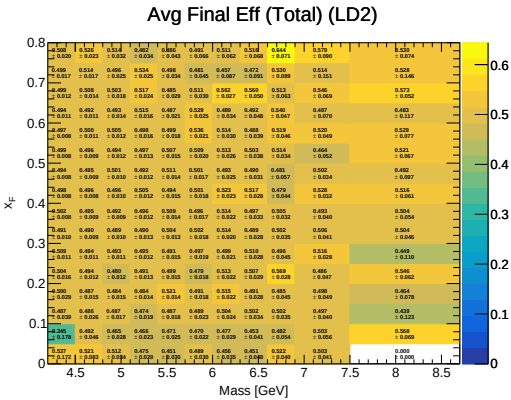
LD2 Target: Total Yield

Final Efficiency Corrections



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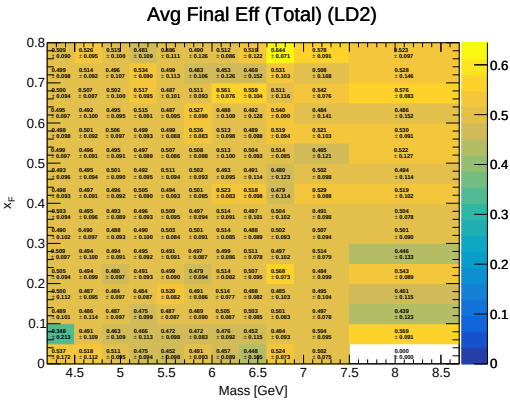

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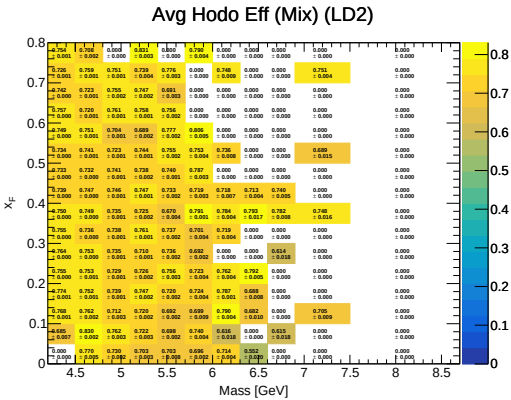
LD2 Target: Mix Yield

Hodoscope Efficiency Corrections



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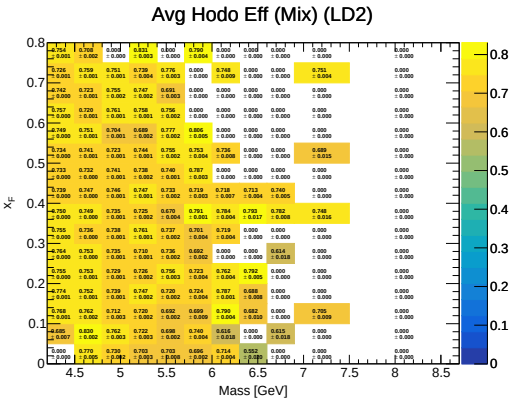

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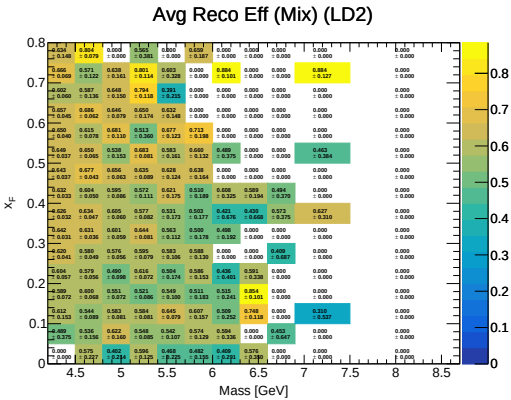
LD2 Target: Mix Yield

Reconstruction Efficiency Corrections



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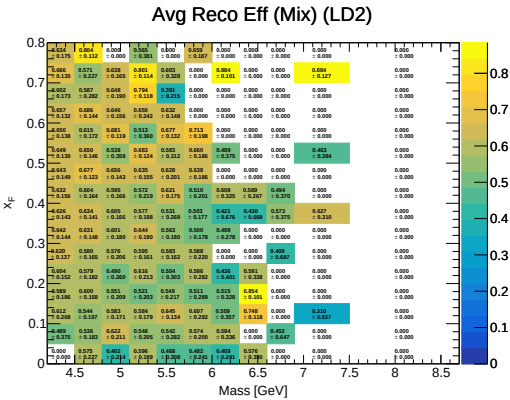

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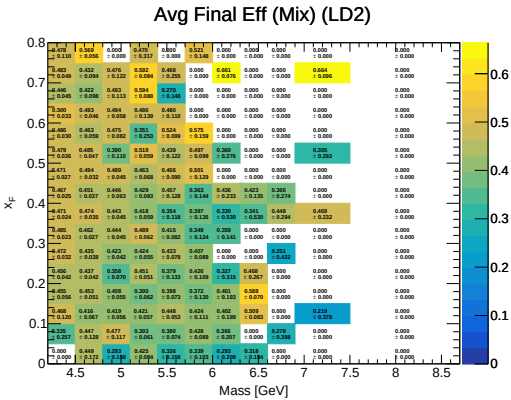
LD2 Target: Mix Yield

Final Efficiency Corrections



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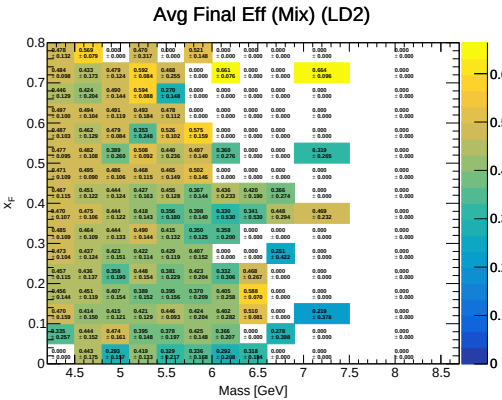
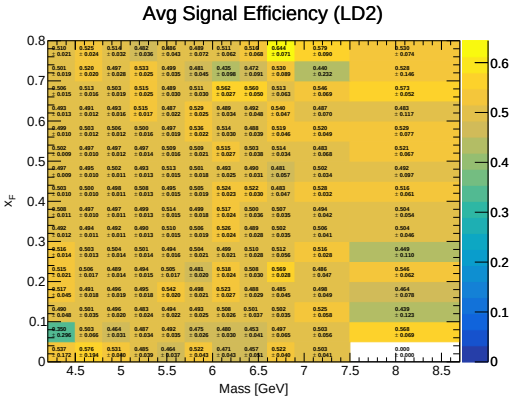


(b) Current

LD2 Target: Signal Efficiency

Final Corrections



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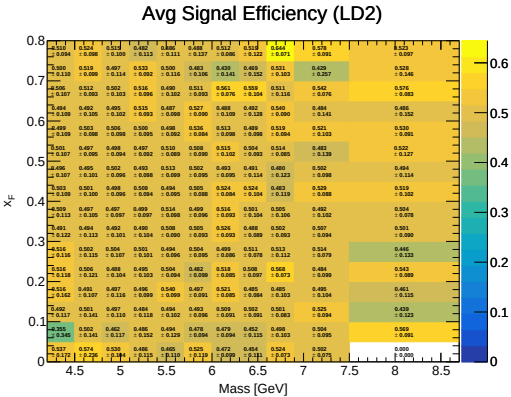
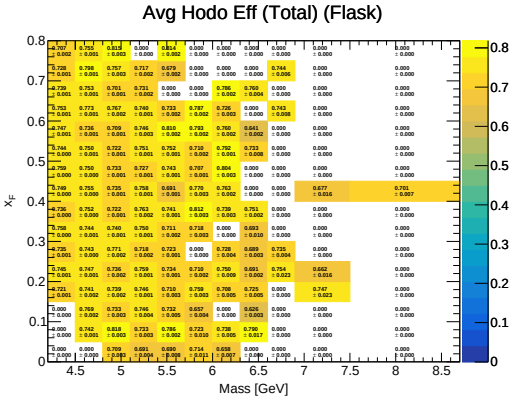


(b) Current

Flask Efficiency Corrections

Flask Target: Total Yield

Hodoscope Efficiency Corrections



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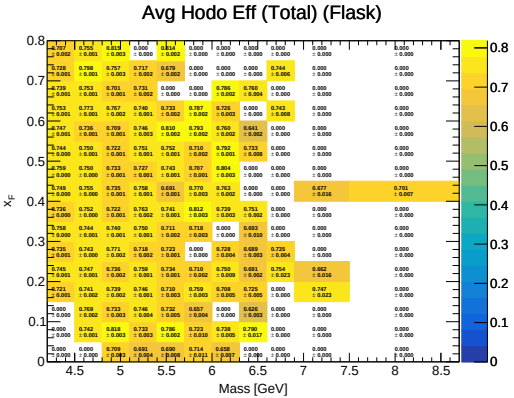

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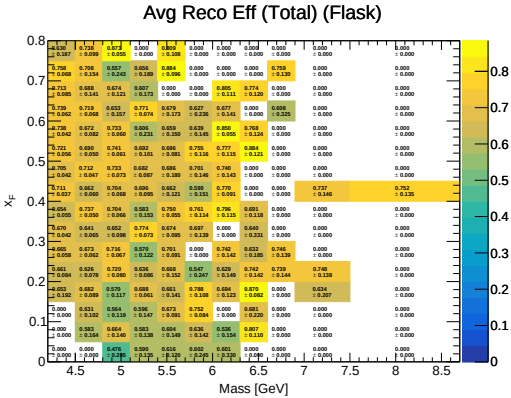
Flask Target: Total Yield

Reconstruction Efficiency Corrections



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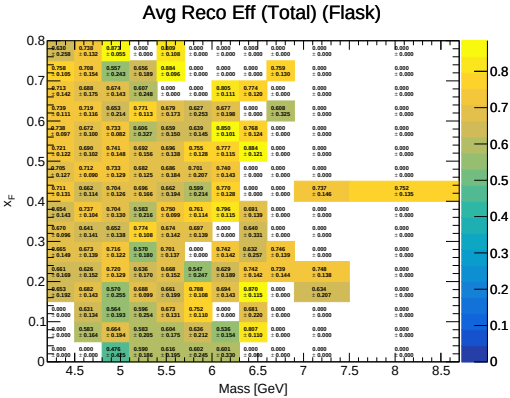
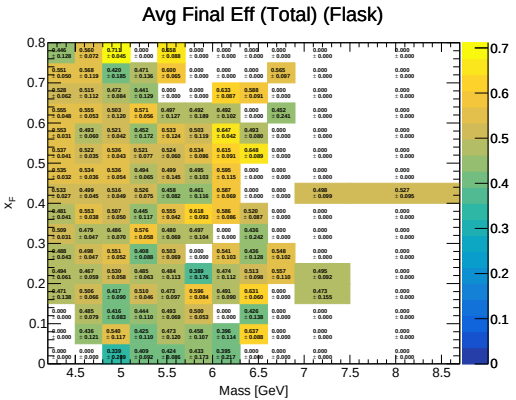


(b) Current

Flask Target: Total Yield

Final Efficiency Corrections



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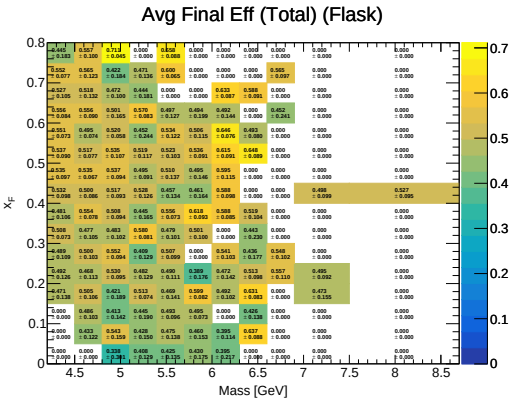
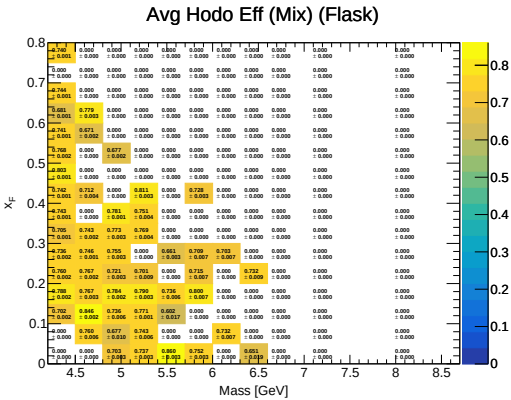


(b) Current

Flask Target: Mix Yield

Hodoscope Efficiency Corrections



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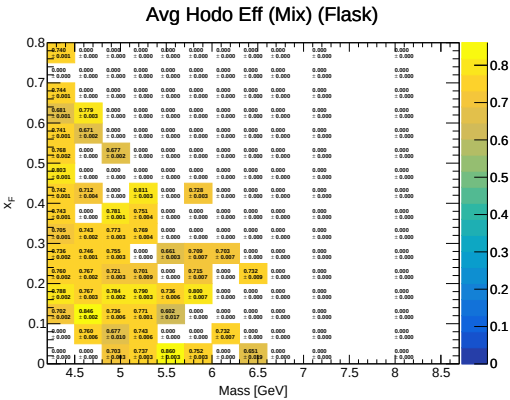

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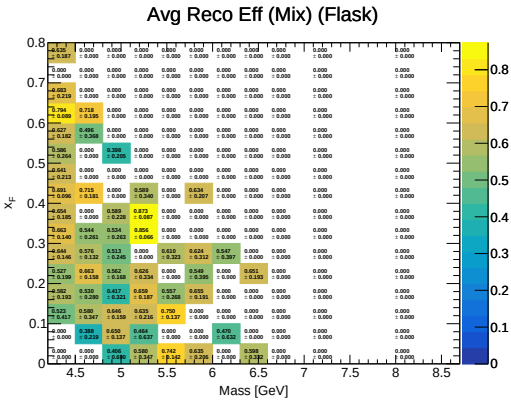
Flask Target: Mix Yield

Reconstruction Efficiency Corrections



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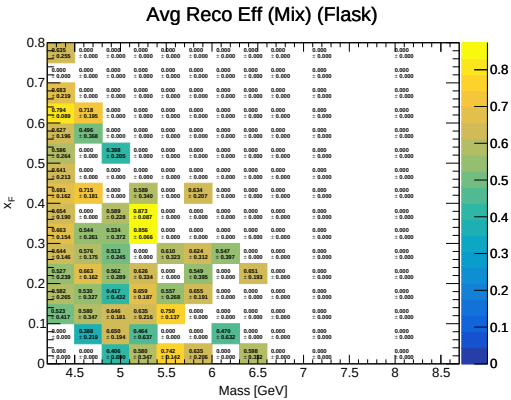
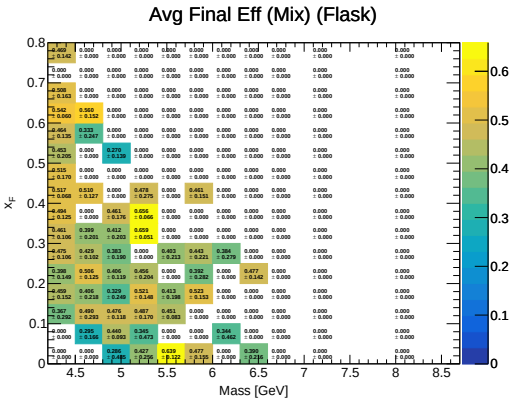


(b) Current

Flask Target: Mix Yield

Final Efficiency Corrections



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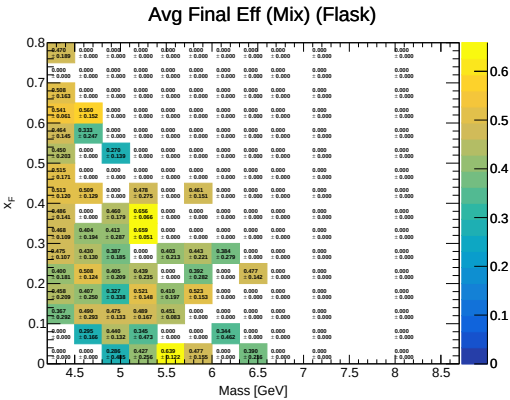
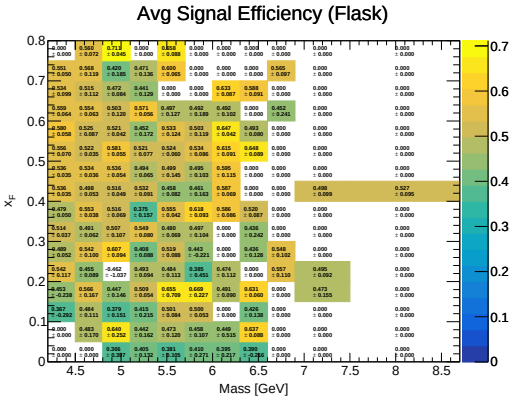


(b) Current

Flask Target: Signal Efficiency

Final Corrections



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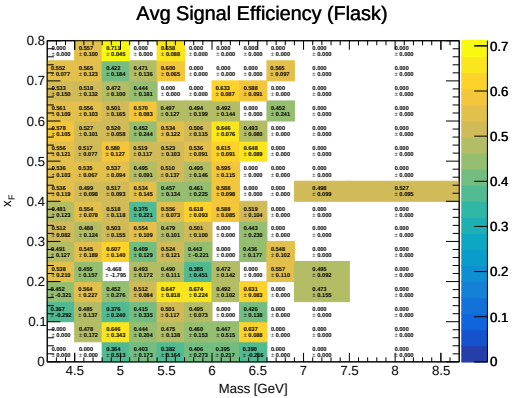

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